

Topic 5: Arms Race (Group 10)

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March 30, 2026



Overview

Scenario: Purple (P) and Green (G) are two countries competing in an arms race.

Research Question: What happens to the long-term spending of each country?

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- How does public sentiment affect defence spending?
- How does limiting spending affect a country's position?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.



Base Model

Model Equations

The equations for the base model are:

$$\begin{aligned}\frac{\partial p}{\partial t} &= ag(t) - bp(t) + r \\ \frac{\partial g}{\partial t} &= cp(t) - dg(t) + s\end{aligned}$$

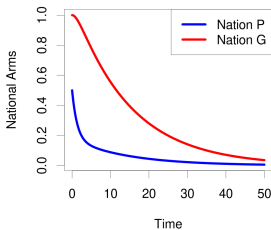
- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c : Mutual Fear ($a, c > 0$)
- b, d : Expenditure ($b, d > 0$)
- $(r, s) > 0$ for trust and $(r, s) < 0$ for hostility

Long Term Behavior

When both countries are neutral, $r = s = 0$, and we have:

$$bd > ac$$

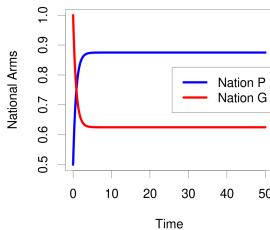
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual
disarmament

$$bd = ac$$

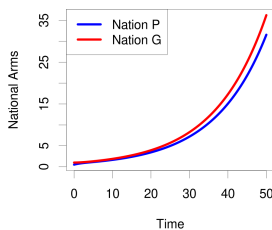
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$

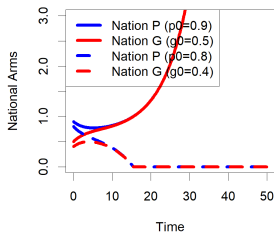


Perpetual arms race

Long Term Behavior

$$bd < ac, r < 0, s < 0$$

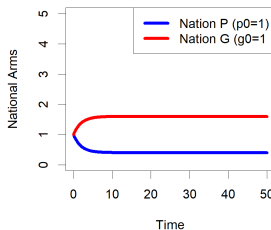
$$a=c=0.25, b=d=0.1, r=s=-0.1$$



Arms Equilibrium
determined by initial
spending

$$bd = ac, r < 0, s > 0$$

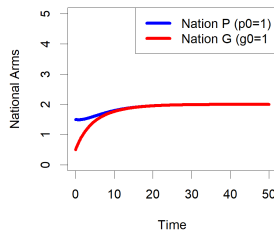
$$a=b=c=d=0.25, r=-0.3, s=0.3$$



Stable Arms spending

$$bd > ac, r > 0, s > 0$$

$$a=c=0.1, b=d=0.25, r=s=0.3$$



Stable Arms spending

Mutual Logistic Fear



Limitation of the Current Model: Each country has a limited amount of resources at a given time.

Solution: Introduce logistic growth.

Model Equations

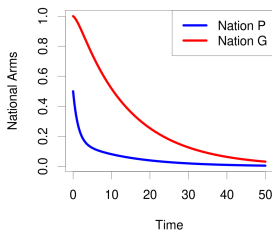
The equations after adding logistic growth to account of limited resources are:

$$\begin{aligned}\frac{\partial p}{\partial t} &= a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r \\ \frac{\partial g}{\partial t} &= c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s\end{aligned}$$

Long Term Behavior

$$bd > ac$$

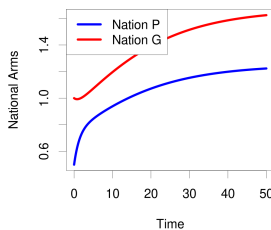
$$a=0.1, b=0.7, c=0.2, d=0.1, K_p=5, K_g=5$$



Complete mutual
disarmament

$$bd = ac$$

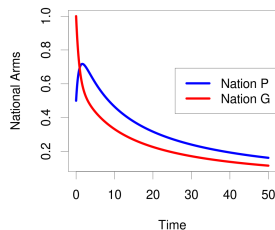
$$a=0.5, b=0.5, c=0.2, d=0.1, K_p=5, K_g=5$$



Stabilized spending

$$bd < ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7, K_p=5, K_g=5$$



Perpetual arms race

Conclusion and Next Steps

Conclusion

- **Base Model:** behaviour based on ratio bd vs ac
- **Public Sentiment:** model behaviour can now change based on initial spending, also impacts equilibrium so stable positive spending is more common
- **Mutual Logistic Fear:** model now accounts for economic limitations, preventing unending arms race to infinite spending

Next Steps (In Report)

- Sensitivity Analysis
- Analytical solutions for equilibrium and stability

Thank you for listening!



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